Chaemin Lim

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Research Interests

Database Systems on Modern Hardware — Concurrent Computing

Publications

1. FaScalSQL: A Fast and Scalable GPU-Accelerated SQL Query Engine for Out-of-Memory Tables
   Chaemin Lim, Suhyun Lee, Jinwoo Choi, Kwanghyun Park, Jinho Lee, Joonsung Kim, and Youngsok Kim
   42nd IEEE International Conference on Data Engineering (ICDE), May 2026
2. DMO-DB: Mitigating the Data Movement Bottlenecks of GPU-Accelerated Relational OLAP
   Chaemin Lim, Suhyun Lee, Jinwoo Choi, Joonsung Kim, Jinho Lee, and Youngsok Kim
   34th International Conference on Parallel Architectures and Compilation Techniques (PACT), Nov. 2025
3. SPID-Join: A Skew-resistant Processing-in-DIMM Join Algorithm Exploiting the Bank- and Rank-level Parallelisms of DIMMs
   Suhyun Lee, **Chaemin Lim**, Jinwoo Choi, Heelim Choi, Chan Lee, Yongjun Park, Kwanghyun Park, Hanjun Kim, and Youngsok Kim
   2025 ACM International Conference on Management of Data (SIGMOD), June 2025
4. PID-Comm: A Fast and Flexible Collective Communication Framework for Commodity Processing-in-DIMMs
   Si Ung Noh[1], Junguk Hong[1], **Chaemin Lim**, Seongyeon Park, Jeehyun Kim, Hanjun Kim, Youngsok Kim, and Jinho Lee
   51st IEEE/ACM International Symposium on Computer Architecture (ISCA), June–July 2024 [1] Co-first authors
5. Design and Analysis of a Processing-in-DIMM Join Algorithm: A Case Study with UPMEM DIMMs
   Chaemin Lim, Suhyun Lee, Jinwoo Choi, Jounghoo Lee, Seongyeon Park, Hanjun Kim, Jinho Lee, and Youngsok Kim
   2023 ACM SIGMOD International Conference on Management of Data (SIGMOD), June 2023
6. GuardiaNN: Fast and Secure On-Device Inference in TrustZone Using Embedded SRAM and Cryptographic Hardware
   Jinwoo Choi[1], Jaeyeon Kim[1], **Chaemin Lim**[1], Suhyun Lee, Jinho Lee, Dokyung Song, and Youngsok Kim
   23rd ACM/IFIP International Middleware Conference (Middleware), Nov. 2022 [1] Co-first authors

Employment

- **System Architect Intern**, Mangoboost — Aug 2025 – Jul 2026
  - Worked on vLLM's GPU inference path for AMD ROCm: wired AMD AITER MLA kernels into multi-token prediction (MTP) decode, fixed a TP8 MTP decode correctness bug, and handled FP8 KV cache dtypes and CUDA graph capture padding.
  - Worked on MoRIIO, vLLM's RDMA-based KV cache transfer for multi-node serving: fixed async KV transfer completion and cleanup, deferred send drain, and READ handoff routing, and added multi-node TP=16 prefill-decode support.
- **Research Assistant**, HPCP Lab., Yonsei University — 2021 – Present

Patents

- **Skew Resistance Processing in a DIMM Device and Processing Method**. US Patent Application 18/933,770, 2026. Youngsok Kim, Suhyun Lee, Chaemin Lim.

- **Electronic Device and Method with Processing in Memory Operation**. US Patent Application 19/060,889, 2025. Bongjun Kim, Youngsok Kim, Sungchul Lee, Seil Lee, Suhyun Lee, Jounghoo Lee, Chaemin Lim, Hanna Cha, Jinwoo Choi, Yeonan Ha, Hanwoong Jung.
- **PIM-Based Computing Device for Accelerating Join Operation**. US Patent 12,386,778, 2025. Youngsok Kim, Chaemin Lim.
- **Artificial Intelligence Device Based on Trust Environment**. US Patent Application 18/229,765, 2024. Youngsok Kim, Jinwoo Choi, Chaemin Lim, Suhyun Lee, Dokyung Song, Jinho Lee.

Invited Talks

- **A Fast Processing-in-DIMM Join Algorithm Exploiting UPMEM DIMMs** @ ABUMPIMP, a minisymposium on applications and benefits of commercial massively parallel PIM platforms, EUROPAR 2023.

Awards

- **Encouragement Prize**, Samsung Humantech Paper Awards, 2024. For "PID-Comm: Fast and Flexible Collective Communication Framework for Commodity Processing-in-DIMM Devices" (Junguk Hong, Si Ung Noh, Seongyeon Park, and Chaemin Lim).

Projects

- **Development of New Concept PIM Semiconductor Leading Technology** Institute for Information and Communications Technology Planning and Evaluation (IITP), 2022
  - Presented UPMEM-based Processing-in-Memory (PIM) research at ABUMPIMP, a minisymposium on applications and benefits of commercial massively parallel PIM platforms.
  - Open-sourced research artifacts for PID-Join (SIGMOD 2023) and PID-Comm (ISCA 2024), enabling reproducible evaluation on commodity UPMEM DIMMs.
  - Developed PIM database and communication systems targeting data movement bottlenecks, join processing, and collective communication on modern memory-centric hardware.
- **STACK: Smart, Attack-Resistant Internet of Things Networks** Korea Institute for Advancement of Technology (KIAT), 2021
  - Reported an STM32MP1 clock-rate computation bug in OP-TEE OS affecting HASH1, GPIOZ, CRYP1, and MDMA clocks; acknowledged by a merged upstream fix with a `Reported-by` tag. OP-TEE/optee_os#4904.
  - Worked with secure embedded-system software stacks, including trusted execution environment (TEE) components for attack-resistant IoT platforms.

Education

- **Yonsei University**, Seoul, Korea — Sep 2023 – Present
  - Ph.D. Student, Computer Science
- **Yonsei University**, Seoul, Korea — Mar 2021 – Aug 2023
  - Master of Science, Artificial Intelligence
- **Hankuk University of Foreign Studies**, Seoul, Korea — Mar 2017 – Feb 2021
  - B.Eng. in Computer Engineering (Double Major)
  - B.A. in Journalism

Teaching Experience

- Logic Circuit Design (CSI2111): Teaching Assistant — Fall 2022

- Open Source Projects (CSI2115): Teaching Assistant — Fall 2021
- Information Security (CSI4109): Teaching Assistant — Spring 2021

Skills

C, C++, Python, Scala, Rust, SQL